

ABU NAKHLAH, Qatar

WMO: 411705

Lat: 25.117N

Lon: 51.315E

Elev: 40

StdP: 100.85

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	10.8	-4.5	2.6	32.8	-2.2	3.2	32.8	12.2	18.0	11.3	18.2	3.5	280	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	46.2	21.3	45.2	21.5	44.3	21.6	29.8	34.6	29.2	34.5	28.7	34.4	6.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.8	25.5	33.1	28.0	24.2	33.2	27.1	23.0	33.1	99.7	34.2	96.7	34.5	93.8	34.3	32.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.6	DB	6.7	48.2	1.8	0.7	5.5	48.6	4.4	49.0	3.4	49.4	2.1	49.9	(4)
(5)			WB	4.2	30.9	2.0	0.7	2.8	31.4	1.6	31.9	0.5	32.3	-1.0	32.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	17.3	19.3	23.2	28.3	33.7	36.0	37.3	37.2	34.4	30.2	24.5	19.3	(6)
(7)		DBStd	7.53	2.66	3.12	2.94	2.95	2.18	1.67	1.31	1.17	1.67	2.11	2.71	2.64	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	98	52	23	2	0	0	0	0	0	0	0	1	21	(9)
(10)		CDD10.0	6732	227	261	409	550	734	781	848	843	732	626	435	288	(10)
(11)		CDD18.3	3787	21	50	152	300	476	531	589	584	482	367	185	50	(11)
(12)		CDH23.3	56726	119	350	1388	3648	7558	9019	10216	9883	7664	4946	1657	277	(12)
(13)	CDH26.7	38651	16	99	596	2043	5202	6636	7737	7403	5308	2909	664	39	(13)	
(14)	Wind	WSAvg	4.2	4.1	4.6	4.8	4.5	5.3	4.8	3.7	3.6	3.3	3.7	4.1	(14)	
(15)	Precipitation	PrecAvg	81	14	11	18	5	1	0	0	0	1	14	17	(15)	
(16)		PrecMax	251	54	69	140	15	11	0	0	0	8	109	90	(16)	
(17)		PrecMin	14	0	0	0	0	0	0	0	0	0	0	0	(17)	
(18)		PrecStd	57	14	15	30	5	3	0	0	0	2	27	26	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	33.0	37.1	41.1	45.9	47.2	47.8	46.9	44.9	41.0	35.9	29.6	(19)
(20)			MCWB	16.1	17.4	17.4	19.1	20.1	21.1	21.9	21.6	21.2	20.7	19.2	17.2	(20)
(21)		2%	DB	26.1	29.6	34.2	39.3	44.1	46.1	46.2	45.8	43.3	39.2	33.8	27.8	(21)
(22)			MCWB	16.2	17.0	17.0	19.0	20.2	20.6	21.7	22.1	22.0	21.3	19.5	17.8	(22)
(23)		5%	DB	24.5	27.1	32.1	37.8	42.9	45.0	45.2	44.8	42.2	38.0	32.1	26.2	(23)
(24)			MCWB	15.8	16.3	16.9	18.8	20.1	20.3	21.9	22.5	22.2	21.0	19.8	17.5	(24)
(25)		10%	DB	22.9	25.2	30.0	35.8	41.2	43.8	44.1	43.8	41.1	36.8	30.6	24.9	(25)
(26)			MCWB	15.2	15.6	16.7	18.4	19.9	20.5	21.9	22.9	22.4	21.6	19.7	17.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.9	21.6	23.1	25.9	27.9	30.1	30.8	29.7	27.3	24.3	22.0	(27)
(28)			MCDB	22.2	23.8	25.3	29.5	31.1	33.8	34.8	34.6	34.6	32.4	28.6	24.4	(28)
(29)		2%	WB	18.7	19.6	20.7	22.2	24.5	26.5	29.3	30.2	28.8	26.5	23.3	20.7	(29)
(30)			MCDB	22.1	22.9	25.5	29.0	31.9	32.9	34.7	34.6	33.8	32.0	28.0	23.5	(30)
(31)		5%	WB	17.7	18.7	19.8	21.3	23.6	25.3	28.7	29.6	28.2	25.7	22.4	19.6	(31)
(32)			MCDB	21.2	22.9	25.3	29.1	32.6	33.8	34.8	34.5	33.3	31.4	27.2	23.1	(32)
(33)		10%	WB	16.9	17.9	18.9	20.7	22.7	24.2	27.8	29.1	27.5	25.1	21.5	18.7	(33)
(34)			MCDB	20.6	22.9	25.2	29.1	32.9	34.0	34.9	34.4	33.2	30.6	26.7	22.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	12.5	13.0	14.9	15.3	14.2	13.9	14.1	13.0	10.6	9.8	(35)
(36)			MCDBR	12.1	13.2	15.7	15.6	16.7	16.9	15.6	15.3	15.3	14.3	12.5	11.3	(36)
(37)			MCWBR	4.9	4.6	4.3	4.3	5.0	5.6	6.4	7.0	6.8	6.2	5.1	4.7	(37)
(38)		5% WB	MCDBR	10.2	10.9	11.7	12.6	15.0	15.7	13.6	13.0	13.6	12.8	11.6	10.0	(38)
(39)			MCWBR	5.2	4.7	4.4	4.6	5.7	6.6	6.3	6.5	6.6	5.9	5.3	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.509	0.557	0.624	0.671	0.665	0.642	0.690	0.647	0.569	0.511	0.507	0.486	(40)	
(41)		taud	1.823	1.716	1.582	1.534	1.530	1.562	1.533	1.617	1.742	1.858	1.870	1.890	(41)	
(42)		Ebn at Noon	723	722	700	682	685	696	663	690	734	753	719	724	(42)	
(43)		Edh at Noon	188	224	269	288	288	278	285	262	226	191	177	169	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.82	4.61	5.33	5.91	6.70	7.05	6.67	6.43	6.00	5.14	4.02	3.60	(44)	
(45)		RadStd	0.23	0.24	0.32	0.41	0.18	0.20	0.18	0.14	0.11	0.12	0.18	0.23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+0.55	N/A	-1.62	+0.70	N/A	N/A	N/A	N/A	+210	+203	(47)

Nomenclature: See separate page